Investigation of the Effect of Incomplete Droplet Prevaporization on NO_x Emissions in LDI Combustion Systems

Christian H. Beck
e-mail: christian.beck@its.uka.de

Rainer Koch

Hans-Jörg Bauer

Institut für Thermische Strömungsmaschinen,
Universität Karlsruhe (TH),
76128 Karlsruhe, Germany

The influence of incomplete liquid fuel prevaporization on the emissions of nitric oxides in a swirl stabilized model gas turbine combustor is investigated experimentally and numerically. The design of the model combustor enables the variation of the degree of prevaporization. This is achieved by using two liquid fuel injectors. One injector is located far upstream of the combustor and generates a fully prevaporized and premixed air fuel mixture. The second injector is located at the combustor inlet. Consequently, the liquid fuel mass flow split between the two injectors determines the fraction of nonprevaporized fuel present in the reaction zone. The NO/NO_2 measurements were performed with a chemoluminescence analyzer. In accordance to the findings of other researchers, the present experimental study revealed that the influence of prevaporization on nitric oxide emissions is of significance for practical applications. The experimental studies were accompanied by numerical studies of partially prevaporized lean combustion in an abstracted configuration. The purpose of this numerical study is to gain a detailed understanding of the influence of droplet slip on droplet flame position and peak temperature. The droplet slip velocity was found to have a significant impact on the peak temperature of the droplet flame and, therefore, NO formation rates within the droplet flame. The combustion system used for the experimental investigation was characterized regarding droplet slip velocities with an extended laser Doppler anemometry technique. The comparison between numerical and experimental results shows that the droplet slip velocities in the macroscopic reaction zone are within the transition range from an envelope to a wake flame. It is concluded that small-scale mixing effects play a significant role in the formation of nitric oxides in spray combustion systems with incomplete prevaporization. [DOI: 10.1115/1.2906185]

Introduction

As of today, the reduction of aero-gas-turbine emissions has become the main driving force for the development of new combustor concepts. The predominant idea is to achieve a reduction of NO_x emissions by minimizing the temporal and spatial inhomogeneities of the temperature in the reaction zone. In lean flames, temperature fluctuations increase NO formation because of the nonlinear dependence of the NO formation on temperature.

Theoretically, a lean perfectly premixed and prevaporized combustion system does offer a significant NO_x emission reduction potential [1]. However, the high combustor pressure levels of modern aeroengines make complete prevaporizing and premixing practically impossible due to the fast onset of autoignition at these conditions [2,3].

The lean direct injection (LDI) concept is characterized by a global lean equivalence ratio and incomplete prevaporization of the liquid fuel. While complete prevaporization is not achieved, large-scale mixture inhomogeneities are avoided. However, the presence of liquid fuel droplets in the macroscopic reaction zone can create mixture inhomogeneities on a microscopic length scale, if a near stoichiometric burning envelope flame is established around individual droplets.

The first question to be addressed in the present work is to what extent the NO_x emissions are affected by the fraction of liquid fuel present in the macroscopic reaction zone. This question was previously investigated by other authors [4,5]. In the first study, a two-injector system was used where the degree of prevaporization is adjusted like in the rig used in the present study. The liquid fuel mass fraction α was varied between $0.3 \leqslant \alpha \leqslant 0$ at $\phi=0.6$; 0.72. The operating pressure was 0.3 MPa and the preheat temperature was 600–700 K. The author reports a piecewise linear dependence of NO_x emissions on α. At higher equivalence ratios, the influence of prevaporization on emissions was found to decrease.

The second study was performed at atmospheric pressure with air preheat temperatures of 323–376 K. The degree of prevaporization was varied by adjusting the premixing duct length and the preheat temperature. The degree of prevaporization was evaluated from phase Doppler (PDA) liquid phase flux measurements. For flame stabilization, an electrically heated wire ring was used. The generally very low preheat temperature causes low adiabatic flame temperatures at lean conditions. The authors report a piecewise linear dependence of emissions on α for $\alpha < 0.5$. For $0.5 < \alpha < 1$, emissions were found not to vary with the degree of prevaporization.

The main difference between the experimental setups of the previous studies and the present one is the type of flame stabilization. In the present study, a generic swirl-stabilized burner is used. The stabilization mechanism is expected to have a significant impact on the flame structure. In addition, the droplet slip is fundamentally different in a swirl-stabilized flame, where gas

Manuscript received November 9, 2007; final manuscript received November 12, 2007. published online June 13, 2008. Review conducted by Dilip R. Ballal.

Copyright © 2008 by ASME

Downloaded 15 Mar 2010 to 129.137.90.23. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

phase velocities significantly change along the droplet trajectories compared to a pipe flow. Therefore, the setup used in the present study can be expected to represent the characteristics of a gas turbine combustor to a greater extent than previous studies.

In the second part of the paper, a detailed analysis of the NO formation in lean flames with partial droplet prevaporization is presented. The linear dependence of NO_x emissions on the degree of prevaporization suggests that the variation of the degree of prevaporization may be abstracted as the superposition of two flame types: droplet flame and premixed flame. The flame type causing high emissions is the individual burning droplet. In order to gain a better understanding of the prevailing NO formation mechanism, the emission formation of a single droplet flame will be numerically studied. It is known that significant droplet slip velocities can cause the stoichiometric envelope flame to extinguish and a wake flame to establish. The influence of this flame type transition on the NO formation of a single droplet flame will be discussed.

First experimental studies on droplet flame extinction were performed by Spalding [6]. Recently, Pope [7] presented a correlation based on these early experiments and new numerical results. The numerical results are based on a setup quite similar to the one used here. The authors report a nonlinear relation between droplet diameter and extinction velocity. Furthermore, the extinction velocity was found to exponentially vary with ambient gas temperature. In general, the study presented by Pope [7] focuses on the point at which extinction of the envelope flame at the forward stagnation point occurs. Investigations using similar numerical setups were performed by Raghavan et al. [8] and Jiang et al. [9]. Both studies focus on the flame type transition caused by the variation of droplet slip velocity. In all available numerical studies on droplet flame extinction, chemical reaction is modeled with a single step global reactions. In the present study, the detailed mechanism from Held et al. [10] is used for several reasons. First, the position of the droplet wake flame is mainly determined by the chemical time scales. Since the droplet flame position was found to have a significant effect on emissions, detailed modeling of the chemical processes is of importance. Second, the formation of nitric oxides is to be modeled. Again, a detailed mechanism from Hewson [11] is used since the accuracy of simplified NO formation rate expressions is limited.

The final experimental investigation presented in this paper will discuss the range of droplet slip velocities found in typical swirl-stabilized combustion systems. By using the results from the numerical investigation, the droplet flame type, which is present in the macroscopic reaction zone of typical swirl-stabilized flames, will be deduced.

Experimental Investigation

Concept and Design of the Experimental Setup. The combustion system used in this study is specifically designed to allow the variation of droplet prevaporization. The setup contains two liquid fuel injectors (see Fig. 1). One injector is positioned far upstream while the second injector is placed in the proximity of the reaction zone. The upstream injector is used to generate a prevaporized and perfectly premixed fuel air mixture, while the downstream injector is used to directly inject droplets into the flame. By varying the fuel mass flow split between the upstream and the downstream injector, the fraction of nonprevaporized fuel present in the reaction zone can be adjusted.

The fraction of liquid fuel present in the flame front is to be determined from fuel mass flow measurements. Therefore, it is necessary that the droplets injected by the downstream injector do not significantly vaporize before they reach the macroscopic reaction zone. This is ensured by injecting the liquid fuel at a temperature of only T_f=263 K. An accompanying numerical study confirmed that an injection temperature of 263 K is sufficient to minimize droplet mass loss due to vaporization for the droplet size

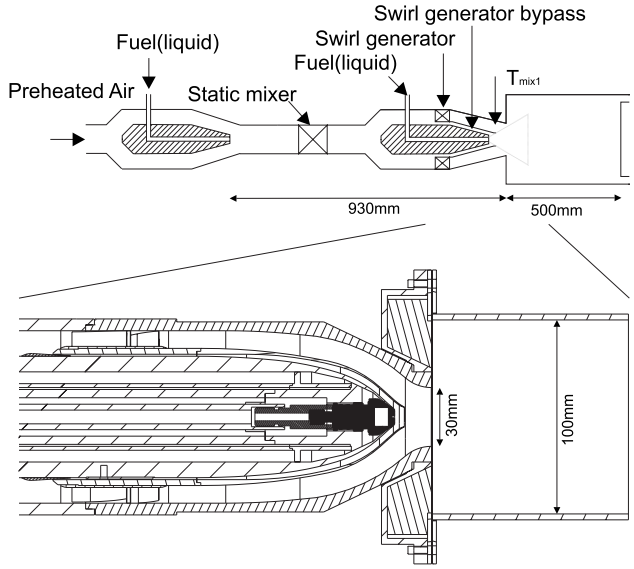

Fig. 1 Schematic of the rig; detail of the burner and combustion chamber

distribution, fuel, air temperature, and residence time of this combustor. Flame stabilization is achieved by generating a vortex breakdown.

Operating Conditions. In the experimental setup, all oxidizer and fuel mass flows are measured using calibrated sensors. The combustor inlet temperature T_{mix1} is controlled as well as the liquid fuel temperature inside the injector. The pressure level behind the combustor exit is atmospheric. An overview of the operating conditions relevant for this paper is given in Table 1.

The combustion system is fed with preheated air. A preheat temperature of T_{mix2}=473 K was chosen. This temperature is a compromise between minimizing liquid fuel prevaporization from the downstream injector and achieving realistic adiabatic flame temperatures (1650–1850 K) at lean conditions ($\phi \leqslant 0.6$). The preheat temperature is defined here as the temperature of the unburnt mixture under the premise that all fuel is vaporized and mixed (T_{mix2}). The combustor inlet temperature T_{mix1} is calculated from this virtual temperature by Eq. (1), in which the temperature change due to heating and vaporization of the nonprevaporized fuel mass flow is accounted for. If T_{mix2} remains constant, the adiabatic flame temperature is independent of the degree of prevaporization α and the liquid fuel temperature. For T_{mix2}=473 K and ϕ=0.5, T_{mix1} linearly varies between 473 K$\leqslant T_{mix1} \leqslant$497 K for $0 \leqslant \alpha \leqslant 1$. The enthalpy of fuel and oxidizer used in Eq. (1) is

Table 1 Operating conditions

Quantity	Value	Comment
ϕ	0.5, …, 0.6	
α	0, …, 0.8	Liquid fuel fraction
T_{mix2}	473 K	
T_f	263 K	Measured in atomizer
$\dot{m}_{Ox}$	14.9 g/s	Total oxidizer mass flow
$\dot{m}_{core}$	0.7 g/s	Swirl generator bypass mass flow
$\dot{m}_{atom}$	0.1, …, 0.25 g/s	Downstream atomizer air mass flow
Fuel	Heptane	
s	0.46	Swirl number at the combustor inlet

Downloaded 15 Mar 2010 to 129.137.90.23. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

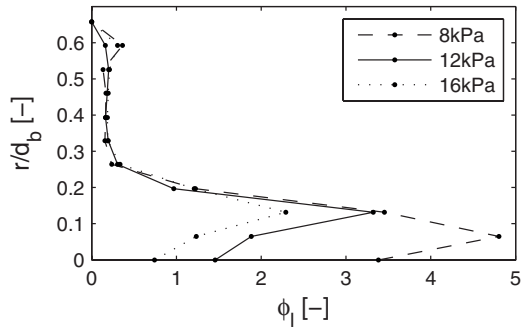

Fig. 2 Radial distribution of liquid fuel equivalence ratio for three atomizer air pressure levels, ϕ=0.55, α=0.5

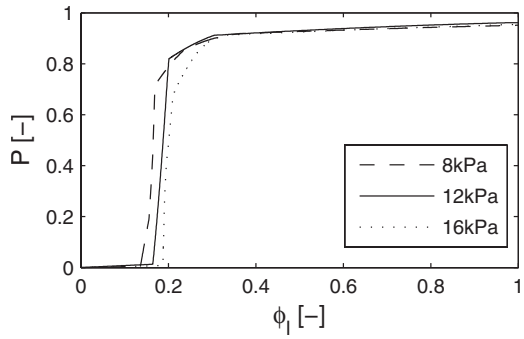

Fig. 3 CDF of liquid fuel equivalence ratio for three atomizer air pressure levels, ϕ=0.55, α=0.5

calculated as a function of temperature. The thermodynamic data provided with the reaction mechanism [10] are used.

$$(h_{\mathrm{air}}+Y_f h_f)|_{T_{\mathrm{mix}\ 2}}=(1-\alpha)Y_f(\Delta h_{lg}+h_f)|_{T_f}+(h_{\mathrm{air}}+\alpha Y_f h_f)|_{T_{\mathrm{mix}\ 1}} \tag{1}$$

Experimental Investigation of Mixture Characteristics. For the injection of liquid fuel, two air assist atomizers are used. The upstream injector was chosen based on the requirement to generate a fine spray suitable for fast prevaporization and premixing. The mixing length is approximately 1 m with a mean residence time of 200 ms. The performance of the premixing system was evaluated in a previous study [12] and was found to deliver a very homogeneous mixture (σ_ϕ=0.008). A traverse mounted gas sampling system and a flame ionization detector were used for the characterization of mixture homogeneity. The measurements were performed 0.4 m downstream of the upstream injector, implying that the standard deviation at the combustor inlet is lower than the value reported above.

For the downstream injector, the key requirements are that the spray characteristics are independent of liquid mass flow, and that the liquid fuel equivalence ratio in the macroscopic reaction zone is as homogeneous as possible. Since these requirements cannot be ideally met, the impact of the deviation from ideal conditions regarding the effects investigated in this study is assessed.

The downstream injector was characterized at reacting conditions with a standard phase Doppler anemometry system setup for a scattering angle of 74 deg. The simultaneous characterization of liquid and gas phase flow was achieved by using the recorded particle diameter as phase discrimination criterion (d_{disc}=3 μm). Since the data rate of small fuel droplets is very low, spherical glass particles with $d<3$ μm and a refractive index of $n\approx 1.5$ were added by a "Palas RGB 1000" particle dispenser. A seeding volume flow was chosen, which is <0.5% of the liquid fuel volume flow. At the chosen scattering angle, the PDA technique is relatively insensitive to particle refractive index [13]. The difference between the expected refractive index of the fuel droplets (n=1.4) and the gas tracer particles causes a sizing error <3%. This sizing error for tracer particles is considered insignificant for the purpose of phase discrimination. The data were sampled at x/d_b=0.33 downstream of the combustor inlet. The radial range covered is $0<r/d_b<0.66$. This location is at the beginning of the macroscopic reaction zone (see Fig. 8).

The radial distribution of liquid fuel equivalence ratio $\phi_l=f_l\rho_l/(c_g\rho_g z)$ was obtained from the PDA measurements (see Fig. 2). The data are shown for an operating condition with 50% prevaporization and ϕ=0.55 at three atomizer air pressure levels. For all atomizer pressure levels, the liquid fuel equivalence ratio is rich near the center axis. The high liquid fuel equivalence ratio in this region is due to the low gas phase velocity near the center axis (see Fig. 9).

In order to assess the importance of the rich region near the center axis, the mixture mass flow weighted cumulative distribution function for the mean liquid fuel equivalence ratio was derived. Mixture mass flow weighting is used since the local mixture mass flow represents the importance of a region for the overall characteristics of the combustion system. The cumulative distribution function (CDF) of the liquid fuel equivalence ratio is shown in Fig. 3. The distribution of the liquid fuel equivalence ratio is relatively narrow in all three cases. Approximately 10% of the mixture mass flow has a liquid fuel equivalence ratio higher than the mean value.

The inhomogeneity of liquid fuel placement could bias the dependence of NO_x emissions on the degree of prevaporization by formation of large-scale temperature inhomogeneities. Since it is not clear to what extent the inhomogeneity of liquid fuel placement affects the overall NO_x emissions of the combustor, a practical test regarding the sensitivity of NO formation on liquid fuel placement was performed. The variation of the atomizing air pressure has a significant effect on the liquid fuel equivalence ratio in the proximity of the burner axis (see Fig. 2). However, the overall NO emissions of the combustion system change by less than 2.5% when a variation of atomizer air pressure is performed (see Table 2). It is concluded that the effect of inhomogeneities in liquid fuel placement, as they are present in the experimental setup, is more than two orders of magnitudes smaller than the effect of the degree of prevaporization on emissions (see Fig. 4). For the emission measurements presented later, an atomizer air pressure level of p_{atom}=12 kPa was chosen. The low sensitivity of NO_x emissions regarding liquid fuel placement can be explained by multiple effects.

1. The mass flow in the region where the inhomogeneities of liquid fuel equivalence ratio are present (burner axis) is small.
2. The length of the droplet trajectories within the macroscopic reaction zone is of the order of the linear diameter. Furthermore, the droplets in the macroscopic reaction zone do not reach momentum equilibrium (see Fig. 9). Therefore, the heat release caused by nonprevaporized fuel is distributed over a large volume. Consequently, the inhomogeneous distribution of liquid fuel at the beginning of the reaction zone does not necessarily imply the existence of a macroscopic

Table 2 Effect of variation of fuel placement on emissions, ϕ =0.55, α=0.5

Case	p_{atom} (kPa)	EI_{NO} (g/kg)	$\bar{D}_{32}$ (μm)
1	8	0.53	52
2	12	0.52	45
3	16	0.52	37

Downloaded 15 Mar 2010 to 129.137.90.23. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

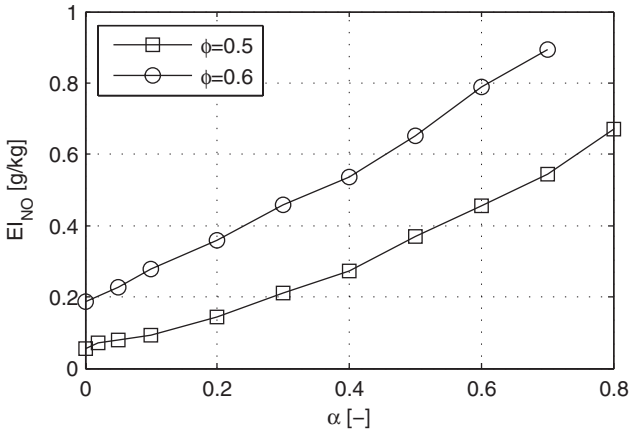

Fig. 4 Influence of liquid fuel mass fraction on overall NO emissions

temperature inhomogeneity and, thus, high NO formation rates.

3. In the present experimental configuration, the gaseous fuel equivalence ratio is not affected by inhomogeneities of the liquid fuel placement, since liquid fuel prevaporization from the downstream injector is minimized by cooling the liquid fuel. An inhomogeneity of the gaseous fuel distribution would cause a macroscopic hot spot and high NO formation rates. Therefore, the impact of liquid fuel placement in the experiment is fundamentally different from real combustors, where the liquid fuel placement has a major impact on gaseous fuel distribution.

An additional source of uncertainty is the dependence of spray properties on the injector liquid mass flow. PDA measurements demonstrated a sensitivity of the liquid mass flow weighted Sauter diameter on the atomizer fuel mass flow. For a variation of the liquid fuel mass fraction between $0.1<\alpha<0.8$, the mean Sauter diameter varied within $40\ \mu\text{m}<\overline{D}_{32}<49\ \mu\text{m}$. Considering the low sensitivity of NO emissions regarding mean droplet diameter (see Table 2), the variation of the mean droplet diameter with the liquid fuel mass fraction is not of significance for the emission measurements to be presented later.

Experimental Investigation of the Influence of Prevaporization on NO_x Emissions. The experimental setup previously described was used to investigate the influence of incomplete prevaporization on the NO_x emissions. The variation of α was performed for two equivalence ratios $\phi=0.5$ and 0.6. The adiabatic flame temperatures corresponding to these equivalence ratios are $T=1650$ K and 1837 K. The NO/NO_x measurements were performed with a chemoluminescence analyzer. Since the measured NO_2 concentrations were virtually zero for all operating conditions, only the NO concentration will be reported.

For both overall equivalence ratios, a significant influence of the liquid fuel fraction α on overall NO emissions is observed (Fig. 4). The relation between NO emissions and the liquid fuel mass fraction is almost linear for both equivalence ratios. This linear dependence of NO emissions on α can be explained by the superposition of two flame types: premixed combustion and droplet burning. Lean premixed flames cause very low emissions of nitric oxides (see Fig. 4, $\alpha=0$) while droplet flames, which may burn at a near stoichiometric equivalence ratio, can cause significant formation of nitric oxides.

Numerical Investigation

Introduction. The experimental results presented in the previous section show that the presence of nonprevaporized droplets in the macroscopic reaction zone has a significant effect on overall combustor NO_x emissions. While the experimental setup allows the variation of global parameters, detailed measurements are extremely difficult to perform and analyze since the quantities of interest show significant fluctuations in time and space due to turbulence. Therefore, spray combustion was also numerically studied. The primary objective of the numerical study is to identify and understand the effects governing the NO formation in partially prevaporized spray flames. An abstracted structure of such a flame is shown in Fig. 5.

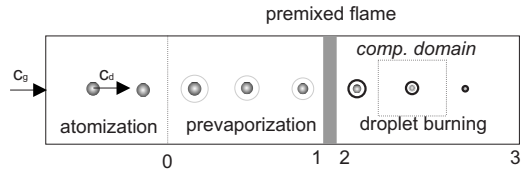

Fig. 5 Abstracted flame structure of partially prevaporized combustion

In the case of a high degree of prevaporization, there will be a stable premixed flame front, which is traversed by droplets. If the degree of prevaporization is reduced, the premixed flame front will extinguish. Then, an alternative flame structure will be formed. This flame structure will still be characterized by an equilibrium of upstream heat transport, which initiates chemical reaction that in turn will cause downstream heat release. A detailed discussion of plausible flame structures is given by Borghi [14].

For a perfectly prevaporized mixture, the temperature of the premixed flame front will be at its maximum value. In contrast, the global NO_x emissions for this case are at a minimum (see Fig. 4). If the degree of prevaporization is reduced, the temperature of the premixed flame front will decrease and cause the region to become less important to overall NO formation. Therefore, the focus of the numerical investigation is on the processes in the proximity of the droplet during droplet burning, which is likely to predominantly contribute to overall emissions.

It is known that the droplet slip velocity influences the burning mode of an individual burning droplet [6]. Two characteristic flame types can be observed: An envelope flame, which encloses the droplet and burns as a diffusion flame at near stoichiometric conditions. At higher slip velocities, the envelope flame extinguishes at the forward stagnation point and a wake flame is established. Since this transition from one flame type to another is expected to be accompanied by changes of the peak flame temperature, NO formation will also be affected by the flame type transition.

For the reasons previously discussed, the effect of droplet slip velocity on flame type transition and NO formation was numerically investigated. The results of the numerical simulation will be discussed considering the droplet slip velocities measured in the presented burner.

Concept of the Numerical Simulation. For the numerical simulation, a configuration representing a partially prevaporized flame burning in a constant area duct with isotropic droplet spacing is considered (Fig. 5). For a given equivalence ratio, degree of prevaporization, oxidizer, and fuel temperature, the local temperature and mixture composition at Position 2, which is the beginning of the computational domain, are calculated. This is done under the assumption that all prevaporized fuel is burnt in the premixed flame and that the mixture has reached chemical equilibrium. From the ratio of the gas and droplet velocity, the local liquid and gas phase density, the overall equivalence ratio, and the degree of prevaporization, the dimensionless droplet spacing l/d can be determined from Eq. (2). The fuel mass fraction Y_f is calculated from the global equivalence ratio.

Downloaded 15 Mar 2010 to 129.137.90.23. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

$$l/d = \left(\frac{(1 - Y_f \alpha) \pi \rho_f \frac{c_d}{c_g}}{6 Y_f \alpha \rho_g} \right)^{1/3} \tag{2}$$

For conditions covered by the experimental investigation, the dimensionless spacing varies between $40 < l/d < 75$. According to Annamalai and Ryan [15], the interaction between burning droplets is small to negligible for this droplet spacing range. Interaction between burning droplets is referred to as group combustion. Various regimes exist, which are generally characterized by the presence of high gaseous fuel concentrations outside flammability limits surrounding a group of droplets in the reaction zone [15]. In lean spray flames with homogeneous fuel placement, such a condition is unlikely to exist since a homogeneous fuel placement implies that the local equivalence ratio is identical to the overall equivalence ratio. Therefore, nonprevaporized droplets will individually burn in LDI combustion systems. Consequently, the simulation does include just one individual droplet. The computational domain is a 2D rectangle, assuming rotational symmetry. The computational domain covers $-20 \leqslant x/d \leqslant 100$ in the streamwise direction and $0 \leqslant r/d \leqslant 37.5$ in the radial direction relative to the droplet center. The domain is initialized with the temperature and mixture composition present at the inlet. An initial droplet temperature of 300 K was chosen. The droplet flame establishes itself due to autoignition. This initial condition is more realistic than initializing an envelope flame [16].

Numerical Modeling. In the present setup, the liquid phase inside the droplet as well as the gas phase is fully discretized. Hence, the modeling of vaporization and interphase heat transfer requires an interface condition. For modeling of hydrocarbon oxidation, a detailed mechanism for n-heptane is used [10]. Additionally, a detailed chemical mechanism developed by Hewson [11] is used to model nitric oxide formation. In the gas phase region, transport equations for momentum, enthalpy, and species mass fractions are solved. The derivation of the transport equation is discussed in detail by Warnatz et al. [17]. In the liquid phase region, only the enthalpy equation is solved.

For the modeling of heat conduction and concentration driven species diffusion, mixture averaged coefficients are used. The thermal conductivity $\bar{\lambda}$ is calculated from the mixture averaged dynamic viscosity assuming a Prandtl number Pr=0.7 with $\bar{\lambda} = \bar{\mu}\bar{c}_p/\mathrm{Pr}$. The diffusion coefficient $\bar{D}$ is calculated from $\bar{D} = \bar{\mu}/(\bar{\rho}\mathrm{Sc})$ with Sc=0.7.

The interface condition is based on the assumption of phase equilibrium at the droplet surface. The vapor pressure required for the coupling of the species transport equation of the fuel was calculated from the Antoine equation (Eq. (3)). The resulting local evaporation flux is needed to set the interface condition for energy and mass conservation (Eq. (4)). The latent heat of vaporization required for the energy interface condition is calculated as a function of temperature (Eq. (5)). The conservation of mass at the interface is technically implemented by locally deforming the grid (Eq. (6)). For spatial discretization, the central differencing scheme was employed. For time stepping, the Crank–Nicolson scheme was used.

$$Y_{f,s} = \frac{M_f}{\bar{M}} \exp\left(A - \left(\frac{B}{T_{g,s} + C} \right) \right) \tag{3}$$

$$c_{f,s,g} = \frac{-\bar{D}\left(\frac{\partial (Y)_f}{\partial (r)} \right)_s}{1 - Y_{f,s}} \tag{4}$$

$$\lambda_{f,l} \frac{dT}{dr} = \rho_{f,g} c_{f,s} h_{l,g}(T_{s,l}) \tag{5}$$

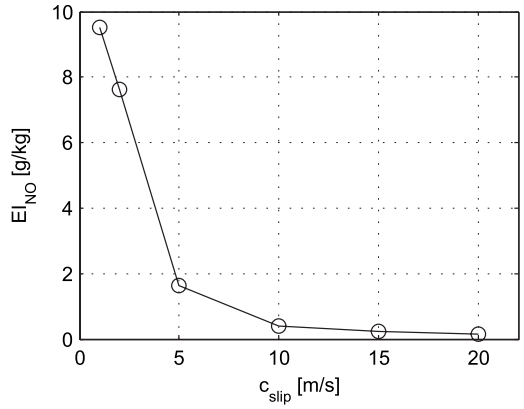

Fig. 6 Dimensionless NO formation rate of single droplet flames

$$c_{f,s,g} \rho_{s,g} Y_{s,f} = c_{f,s,l} \rho_{f,s,l} \tag{6}$$

Results. A variation of the droplet slip velocity was performed for one operating condition characterized by the preheat temperature T_{mix2}=673 K, liquid fuel fraction α=0.1, global equivalence ratio ϕ=0.5, and p_{amb}=0.1 MPa. For these boundary conditions, the temperature downstream of the premixed flame and at the beginning of the computational domain is 1805 K. The initial droplet diameter is d=50 μm. The droplet slip velocity was varied between 1 m/s $< c_{\mathrm{slip}} <$ 20 m/s, which corresponds to a droplet Reynolds number of $0.18 < \mathrm{Re}_D < 3.6$.

Since the simulation is transient, a time for comparison of the nonintegral quantities needs to be chosen. For the operating conditions considered here, the droplet flames have reached a stable burning mode after t=85,...,350 μs, depending on the flame type. The comparison was performed at a simulation time t=500 μs.

For the investigation of the influence of the slip velocity on NO formation, the ratio of the NO formation rate integrated over the domain volume (Eq. (7)) and the vaporization fuel mass flow of the droplet is used to quantify NO formation. In a global context, this is referred to as the emission index EI_{NO}.

$$\dot{m}_{\mathrm{NO}} = M_{\mathrm{NO}} \int \sum_{i=1}^{m} z_i R_i dV \tag{7}$$

In Fig. 6, the simulated effect of droplet slip on NO formation in the droplet flame is shown. For the range of droplet slip velocities considered, a variation of NO formation of almost two orders of magnitude was found. The relation between emissions and slip velocity exhibits a nonlinear transition in the slip velocity range of 2 m/s $< c_{\mathrm{slip}} <$ 5 m/s. An explanation of this behavior can be obtained from a comparison of the flame structure.

The predicted temperature field for three different droplet slip velocities is shown in Fig. 7. The temperature field around the droplet is significantly affected by droplet slip. Toward higher slip velocities, it is becoming more homogeneous, while a decrease in peak temperature is observed. Additionally, the region of the temperature peak moves further away from the droplet.

From detailed analysis of the temperature and CH radical mass fraction fields, it was concluded that the mass fraction peak of the CH radical correlates quite well with the location of the heat release (Fig. 7). For the smallest slip velocity of 2 m/s, an envelope flame is observed. However, the temperature and CH mass fraction at the forward stagnation point are lower than in the wake region. In the case of a slip velocity of 5 m/s, no significant heat release is observed in the upstream region of the droplet as the local temperature does not exceed the inlet temperature. Therefore, the extinction of the droplet flame at the forward stagnation

Downloaded 15 Mar 2010 to 129.137.90.23. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

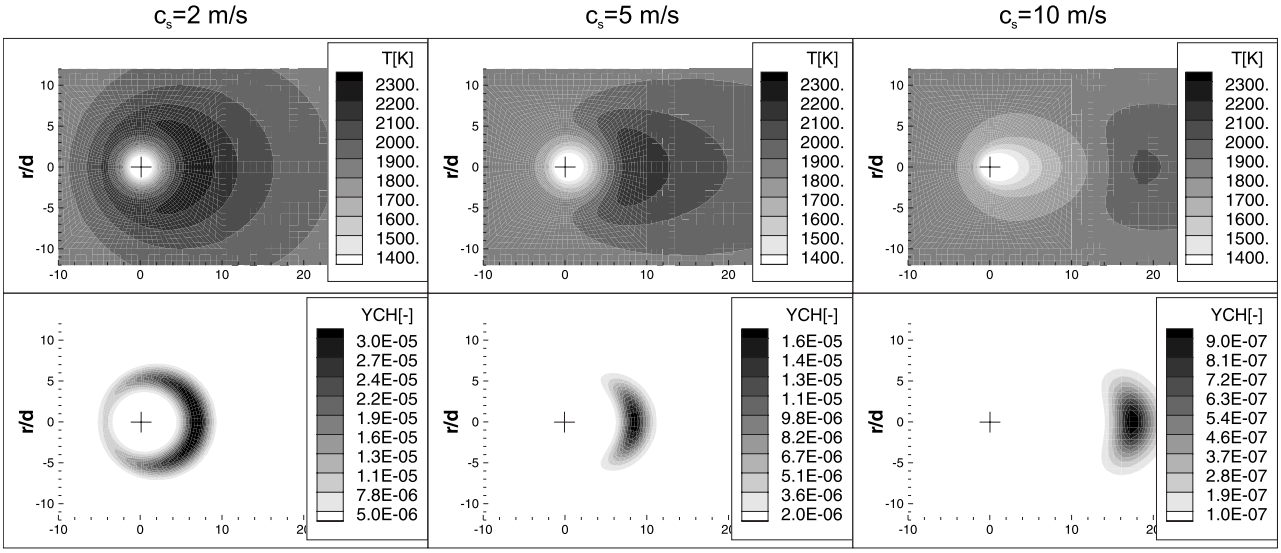

Fig. 7 Effect of slip velocity on temperature and CH mass fraction

point must occur for a slip velocity of $2,\ldots,5$ m/s. If the slip velocity is further increased, the flame position moves further downstream. The transition between the two droplet flame types explains the nonlinearity in the relation between droplet slip and NO formation, as shown in Fig. 6.

Considering the results of the numerical investigation, the effect of droplet slip on NO formation in a droplet flame can be explained by the following mechanism.

1. The flame position is determined by a chemical time scale and a convective time scale. An increase of the slip velocity results in a decrease of the convective time scale and, consequently, the distance between droplet and flame increases.
2. The distance between droplet and flame exerts a significant influence on the equivalence ratio of the droplet flame. In the case of a large distance, mixing is improved, resulting in a lower equivalence ratio of the droplet flame.
3. The equivalence ratio at the position of the droplet flame determines the flame temperature. Due to the nonlinear dependence of NO formation on temperature, the equivalence ratio has a significant impact on NO formation in the flame.

In the present analysis, radiative heat transfer was not considered. Radiative heat transfer can reduce the peak temperature of the droplet flame. Furthermore, the vaporization rate of the droplet can be increased if the droplet absorbs radiation. The effect of thermal radiation is, however, not expected to qualitatively alter the effect of slip velocity on NO formation in droplet flames.

Experimental Investigation of Droplet Slip Velocities

In the previous section, it was shown that the droplet slip velocity has a strong impact on the microscopic droplet flame shape as well as local peak temperature, which in turn significantly affects NO formation. In the experimental investigation, it was found that the presence of liquid fuel in the macroscopic reaction zone has a significant effect on overall emissions. In order to show which magnitude of droplet slip velocities and, consequently, droplet flame type can be expected in a swirl-stabilized burner, the droplet slip velocity was measured in the generic burner. As a prerequisite, the location of the macroscopic reaction zone had to be identified. Then, the mean gas phase and droplet velocities were acquired in this region.

Identification of the Macroscopic Reaction Zone. In order to determine the mean slip velocities of the burning droplets, first the region where there is a high probability of the presence of burning droplets needs to be identified. The recording of time averaged chemiluminescence intensity and subsequent deconvolution is a technique well suited to obtain this information. For the deconvolution, the Abel transformation is used.

In the deconvoluted chemiluminescence recordings, two separate regions are visible where significant chemiluminescence intensities are observed (see Fig. 8). The main flame burns in the shear layer between fresh mixture and recirculated mixture. The second region is located near the center axis in the inner recirculation zone. The extent of the inner recirculation zone and the position of the shear layer is characterized by the contour plot of the axial gas phase velocity shown in Fig. 9.

Measurement of Droplet Slip in the Reaction Zone. For the determination of droplet slip velocities, the disperse phase as well as the gas phase velocities were simultaneously measured. An LDA setup in combination with a phase discrimination technique is used. Phase discrimination is performed by particle diameter. The particle diameter is obtained from the recorded scattering intensity. A discrimination diameter of $d=2$ μm was chosen. The effect of Gaussian laser intensity distribution in the measurement volume on the scattering intensity is reduced by rejecting particles, which do not traverse the near center region of the measurement volume. This is achieved by calculating the particle trajectory length within the measurement volume and defining a boundary value $l_{\text{bound}}=0.8d_{\text{probevol}}$. The results of the extended LDA technique are in good agreement with disperse phase veloc-

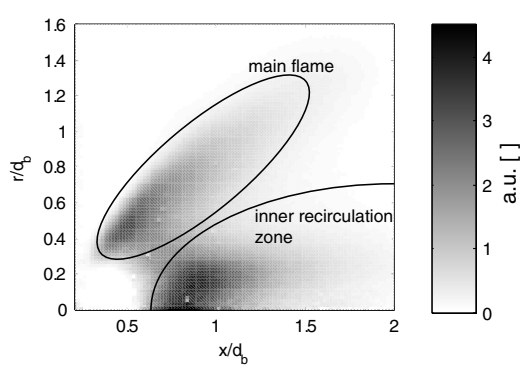

Fig. 8 Chemiluminescence intensity, time averaged and deconvoluted, band pass filtered for CH* emission around 431 nm; $\phi=0.5$, $\alpha=0.5$

Downloaded 15 Mar 2010 to 129.137.90.23. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

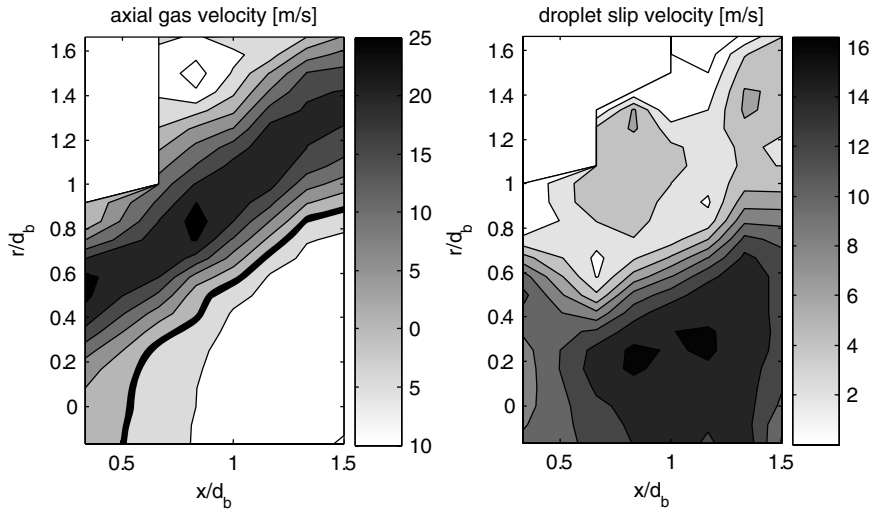

Fig. 9 Axial gas phase velocity and droplet slip velocity magnitude, ϕ=0.5, α =0.5, thick black line: c_{ax}=0 m/s

ity measurements obtained with the PDA technique. The extended LDA technique was preferred over the PDA technique for these measurements since the optical access window available for LDA was less susceptible to degradation by tracer particles.

The mean value of the axial and tangential velocity component was obtained for both phases. The resulting droplet slip velocity is shown in Fig. 9. The highest droplet slip velocities of $12\ \mathrm{m/s} < c_{\mathrm{slip}} < 16\ \mathrm{m/s}$ are observed near the center axis, where the inner recirculation zone is located. Further outside, there exists a region with lower slip velocities of $2\ \mathrm{m/s} < c_{\mathrm{slip}} < 5\ \mathrm{m/s}$. This is the region of the main heat release. Thus, two regions characterized by different droplet combustion modes can be discriminated.

1. The inner recirculation zone, where high droplet slip velocities prevail. According to the numerical study previously presented (Fig. 7) at high slip velocities, a wake flame burning behind the droplet is expected to exist. The wake flame is characterized by low flame temperatures and, thus, a minor contribution to the overall NO_x emissions.
2. The main flame, where low droplet slip velocities are observed. In this region, the numerical simulation predicts the existence of both envelope and wake flames since the slip velocities are within the range of the flame type transition. As a consequence, the droplet flame temperature is higher, which results in higher NO formation rates.

Summary

The experimental study shows that the degree of prevaporization has a significant impact on the NO_x emissions of a swirl-stabilized model gas turbine combustor. Thus, the reduction of nonprevaporized fuel in the reaction zone does offer a significant NO_x emission reduction potential for LDI combustion systems. With increasing overall equivalence ratio, the emission reduction potential decreases. The effect of the degree of prevaporization on the NO_x emissions is a consequence of the presence of droplets in the macroscopic reaction zone. The droplet density in the reaction zone is in the range in which individual droplet burning is expected. The absence of group combustion phenomena is a general feature of LDI combustion systems with homogeneous fuel placement.

An almost linear relation between the degree of prevaporization and the overall nitric oxide emissions was found. The linearity suggests that a superposition of two NO formation mechanisms exists. While nitric oxide formation in lean premixed flames is minimal, a near stoichiometric droplet flame can be a significant source of nitric oxides. Therefore, the amount of nonprevaporized fuel which is burnt in droplet flames is directly proportional to the overall NO_x emissions.

As a major result of the present study, an alternative method for the NO_x emission reduction in LDI combustion systems was identified. NO formation can be minimized for a given degree of prevaporization by maximizing the droplet slip velocity in the macroscopic reaction zone. The effect of the droplet slip velocity on NO_x emissions is governed by the following mechanism: An increase in droplet slip velocity will increase the droplet/flame distance, and due to an increase of the mixing length, the mixture fraction at the flame position is reduced. Consequently, the peak temperature as well as the NO formation rates are lowered.

The mean droplet slip velocities in the swirl-stabilized model combustor investigated in the present study were measured. In the main reaction zone, relatively low droplet slip velocities were measured, which are within the transition range between envelope and wake flame. In the inner recirculation zone, relatively high slip velocities were measured, suggesting the presence of wake flames. Thus, it is concluded that major contributions to the NO_x emissions under nonprevaporized conditions stem from droplets burning in the main reaction zone.

The final conclusion of the present study is that a reliable prediction of the NO_x emissions of LDI combustors must include a submodel that accounts for single droplet burning and the nitric oxides formed within these droplet flames.

Acknowledgment

The authors gratefully acknowledge the financial support for this project by the European community within the framework of the research program TLC "Towards Lean Combustion" (Contract No. AST4-CT-2005-012326).

Nomenclature

Latin Letters

c = velocity magnitude (m/s)
D = diffusion coefficient ($\mathrm{m^2/s}$)
d = diameter (m)
D_{32} = Sauter mean diameter (SMD) (m)
f = flux ($\mathrm{m^3/(m^2\,s)}$)
h = total enthalpy (J/(kg K))
l = droplet spacing (m)
M = molar mass (kg/mol)

Downloaded 15 Mar 2010 to 129.137.90.23. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

$\dot{m}$ = mass flow (kg/s)
R = species source term (kg/s)
s = swirl number
T = temperature (K)
u = velocity component (m/s)
X = molar fraction
Y = mass fraction
z = stoichiometric factor

Greek Letters

α = ratio of liquid to total fuel mass
ϕ = equivalence ratio
λ = thermal conductivity (W/(m K))
ρ = density (kg/m^3)

Subscripts

ax = axial
b = burner
f = fuel
g = gas phase
l = liquid phase
mix1 = condition after mixing of prevaporized fuel fraction
mix 2 = virtual condition after complete vaporization
ox = oxidizer
r = radial
s = droplet surface

References

[1] Bauer, H.-J., 2004, "New Low Emission Strategies and Combustor Designs for Civil Aeroengine Applications," Prog. Comput. Fluid Dyn., **4**, pp. 130–142.
[2] Aggarwal, S. K., 1998, "A Review of Spray Ignition Phenomena, Present Status and Future Research," Prog. Energy Combust. Sci., **24**, pp. 565–600.
[3] Cano-Wolff, M., Meisl, J., Koch, R., and Wittig, S., 1998, "The Influence of Evaporation on the Autoignition-Delay of n-Heptane Air Mixtures Under Gas Turbine Conditions," *27th International Symposium on Combustion*, pp. 2025–2031.
[4] Cooper, L. P., 1980, "Effect of the Degree of Prevaporization Upon Emissions for a Premixed Partially Vaporized Combustion System," NASA Technical Paper No. 1582.
[5] Baessler, S., Moesl, K. G., and Sattelmayer, T., 2006, "NO_x Emissions of a Premixed Partially Vaporized Kerosene Spray Flame," ASME Paper No. GT2006-90248.
[6] Spalding, D. B., 1953, "The Combustion of Liquid Fuels," *Fourth International Symposium on Combustion*, pp. 847–864.
[7] Pope, D. N., and Gogos, G., 2005, "Numerical Simulation of Fuel Droplet Extinction Due to Forced Convection," Combust. Flame, **142**, pp. 89–106.
[8] Raghavan, V., Babu, V., Sundararajan, T., and Natarajan, R., 2005, "Flame Shapes and Burning Rates of Spherical Fuel Particles in a Mixed Convective Environment," Int. J. Heat Mass Transfer, **48**, pp. 5354–5370.
[9] Jiang, T. L., Chen, W. S., Tsai, M. J., and Chiu, H. H., 1995, "A Numerical Investigation of Multiple Flame Configurations in Convective Droplet Gasification," Combust. Flame, **103**, pp. 221–238.
[10] Held, T. J., Marchese, A. J., and Dryer, F. L., 1997, "A Semi-Empirical Reaction Mechanism for n-Heptane Oxidation and Pyrolysis," Combust. Flame, **123**, pp. 107–146.
[11] Hewson, J. C., 1997, "Pollutant Emissions From Nonpremixed Hydrocarbon Flames," University of California.
[12] Schaefer, O., 2005, "Experimentelle und Numerische Analyse des Flammenrueckschlags bei der Mageren Vormischverbrennung," ITS, University of Karlsruhe.
[13] Maier, G., and Wittig, S., 1999, "Fuel preparation and Emission Characteristics of a Pressure Loaded LPP Combustor," AIAA Paper No. 99-3774.
[14] Borghi, R., 1996, "Background on Droplets and Sprays," Lecture Series-von Karman Institute, Vol. 2, pp. 1–39.
[15] Annamalai, K., and Ryan, W., 1992, "Interactive Processes in Gasification and Combustion. Part 1: Liquid Drop Arrays and Clouds," Prog. Energy Combust. Sci., **18**, pp. 221–295.
[16] Chiu, H. H., 1996, "Multiple-State Phenomena and Hysteresis of a Combusting Isolated Droplet," Atomization Sprays, **6**, pp. 1–26.
[17] Warnatz, J., Maas, U., and Dibble, R. W., 2001, *Combustion*, Springer, New York.

Downloaded 15 Mar 2010 to 129.137.90.23. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm